VARUN SURESH

Health Data & Bioinformatics Analyst · Machine Learning

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linkedin.com/in/varun-suresh · github.com/your-github-username · **Eligible to work in the UK** (Graduate Route visa)

PROFILE

First-Class Data Science & Human Biology graduate who works across the rare intersection of **machine learning and the biology behind the data**. I build end-to-end pipelines on real clinical datasets — a 100,000-record diabetes model at 90%+ accuracy, RNA-seq differential-expression analysis, and microbiome cancer-biomarker research — and translate results for clinical, non-technical audiences. Strong in Python, SQL and statistical analysis, with hands-on bioinformatics experience and a GDPR/NHS-aware approach to health data.

TECHNICAL SKILLS

Languages & Data: Python (pandas, NumPy, scikit-learn, XGBoost, matplotlib), SQL, Java, R (basic), HTML/CSS, JavaScript, PHP

Data & ML: Data cleaning & EDA, feature engineering, classification & regression, model evaluation (ROC/AUC, precision/recall), cross-validation, hyperparameter tuning, end-to-end pipelines, data visualisation, dashboarding

Bioinformatics & Health Data: RNA-seq & transcriptomics, differential expression analysis, microbiome analysis, biomarker discovery, Galaxy platform, clinical-data interpretation

Tools & Practice: Git/GitHub, Jupyter, VS Code, relational databases | GDPR & NHS data-governance principles

EDUCATION

BSc (Hons) Data Science & Human Biology — Keele University 2023 – 2026

First-Class Honours · multiple first-class modules

- **Dissertation 1 — Applied ML:** machine-learning pipeline for early diabetes detection across 100,000 health records; 51 engineered features, 8 benchmarked models, 90%+ accuracy & AUC.
- **Dissertation 2 — Health Data Research:** blood-microbiome biomarkers for cancer detection, linking microbial signals to clinical outcomes.
- **Key modules:** Artificial Intelligence, Machine Learning, Data Visualisation, Omics Technologies, Research Methods, Cell Biology, Physiology.

KEY PROJECTS

Early Detection of Diabetes — End-to-End ML Pipeline 2025 – 2026

- Designed and built a full Python ML pipeline over **100,000 real-world health records**, from raw data to evaluated models.
- Engineered **51 clinically informed features** from 33 raw variables, using domain knowledge to boost predictive signal.
- Trained and benchmarked **8 classifiers** (Logistic Regression, Random Forest, XGBoost, SVM, KNN, Decision Tree, Naïve Bayes, Neural Network), reaching **90%+ accuracy and AUC**.
- Applied cross-validation, hyperparameter tuning and error analysis; documented results in a reproducible report. *(Python, scikit-learn, XGBoost, pandas)*

Transcriptomic Analysis — Multi-Gene Expression Study 2025

- Ran the full RNA-seq workflow on an 11-gene dataset (QC, read alignment, differential expression, statistical visualisation) and interpreted findings for non-specialist audiences. *(Galaxy, RNA-seq, DEA)*

Blood Microbiome as a Cancer Biomarker 2025

- Identified microbial biomarkers separating cancer from non-cancer samples via statistical analysis, contextualised within current oncology literature. *(Python, statistical analysis)*

UK Schools Dataset Analysis & SQL Database System 2024

- Cleaned, analysed and visualised a **9,000-school** national dataset, then designed a fully normalised relational SQL database with complex multi-table queries. *(Python, SQL)*

Full-Stack Web App & Java Banking System 2023 – 2024

- Built a secure multi-user web app (auth, dashboard, live weather API) and an object-oriented Java banking system with full transaction and error handling. *(PHP, JavaScript, Java/OOP)*

EXPERIENCE

Front Desk Host & Receptionist — Alton Towers Resort Feb 2025 – Present

- Raised guest-satisfaction ratings from **2.9 to 4.6** through proactive issue resolution; handled high-volume bookings, payments and cross-team communication under pressure.

VR Assistant — FUNLAND Indoor Playground | **Administrative Assistant** — Sri Value Tech 2022 – 2024

- Maintained technical equipment and troubleshooting under load (FUNLAND); handled data entry, documentation and operational reporting (Sri Value Tech).

ADDITIONAL

- **Languages:** English (fluent), Hindi & Kannada (native), Tamil & Telugu (conversational), German (elementary). **Member:** Keele Data Science Club.